

Table 1: Examples of conditions that could be either traumatic and non-traumatic as an origin

Terms indexed to traumatic categories	Indexed to:	Possible opposite origin:
Subdural hematoma	NA07.6Z Traumatic subdural haemorrhage, unspecified whether acute or chronic	8B02 Nontraumatic subdural haemorrhage
Subdural haemorrhage	NA07.6Z Traumatic subdural haemorrhage, unspecified whether acute or chronic	8B02 Nontraumatic subdural haemorrhage
Laceration of heart	NB31.11 Laceration of heart without haemopericardium	BC4Y Other specified diseases of the myocardium or cardiac chambers
Hypothermia	NF02 Hypothermia	MG28 Hypothermia, not associated with low environmental temperature
Asphyxiation	NF05 Asphyxiation	MD11.1 Asphyxia

Terms indexed to nontraumatic categories	Indexed to:	Possible opposite origin:
Meningitis	1D01.Z Infectious meningitis, unspecified	NF0A.Y Other early complication of trauma, not elsewhere classified
Cerebral hemorrhage	8B00.Z Intracerebral haemorrhage, site unspecified	NA07.1 Traumatic intracerebral haemorrhage
Emphysema	CA21.Z Emphysema, unspecified	NF0A.7 Traumatic subcutaneous emphysema, not elsewhere classified
Pneumothorax	CB21.Z Pneumothorax, unspecified	NB32.0 Traumatic pneumothorax
Asphyxia	MD11.1 Asphyxia	NB32.0 Traumatic pneumothorax
Hypovolaemic shock	MG40.1 Hypovolaemic shock	NF0A.4 Traumatic shock, not elsewhere classified

Note that the indexing may be affected by translation (e.g. asphyxiation/asphyxia).

Table 2: List of manner of death and the origin indicated

Manner of death	Indicated origin
Disease	Nontraumatic
Accident	Traumatic
Intentional self-harm	Traumatic
Assault	Traumatic
Legal intervention	Traumatic
War	Traumatic
Could not be determined	Traumatic*
Pending Investigation	Do not indicate its origin
Unknown (not selected)	Do not indicate its origin

Note*: While ‘undetermined intent’ in Chapter 23 is understood as traumatic (See 2.21.8.8 Intent of external causes), the interpretation of the manner of death ‘could not be determined’ may differ depending on different national practice that should be taken into account. For example, there may be countries where it may not indicate any specific origin.

1. When the condition is reported as due to a nontraumatic condition (codable to chapters other than Chapter 22 or 23)

Consider the condition as nontraumatic, regardless of the manner of death reported, and code to the nontraumatic code.

Example 1

- 1
 - (a) Laceration of heart {[NB31.11]} [BC4Y](#)
 - (b) Myocardial infarction [BA41.Z](#)
 - (c)
 - (d)

2

Manner of death: Disease

Laceration of heart is indexed to [NB31.11](#) which indicates a traumatic origin. There is another condition ‘Myocardial infarction’ indexed to a nontraumatic origin, reported as causing laceration of heart. Consider the condition as nontraumatic, and code to [BC4Y](#).

2. When the condition is reported as due to a traumatic condition (codable to Chapter 22 or 23)

Consider the condition as traumatic, regardless of the manner of death reported, and code to the traumatic code.

Example 2

- 1 (a) Subdural hematoma [NA07.6Z](#)
(b) Fell while walking [PA6Z](#)
(c)
(d)

2

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic*, which indicates a traumatic origin. 'Fell while walking' reported as the cause of subdural hematoma, indicating it was traumatic. Keep the code [NA07.6Z](#) as indexed, even though the manner of death was selected as "Disease".

3. When no condition is reported as its cause (reported as such or in a sequence), and:

3a) if an external cause is mentioned either in Part 1 or Part 2, or how the external cause occurred is described in Frame B:

consider the condition as traumatic, regardless of the manner of death reported, and code to the traumatic code. Note: Do not apply this when there are other information indicating the condition is unrelated to the external cause stated. For example, the duration of the condition and the date of injury indicates that they were separate events.

Example 3

- 1 (a) Acute heart failure [BD1Z](#) & [XT5R](#)
(b) Subdural hematoma [NA07.6Z](#)
(c)
(d)
- 2 Fell while walking, dementia [PA6Z](#), [6D8Z]

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) which indicates a traumatic origin. There is no cause reported in a sequence that would indicate its origin, however, an external cause is reported in Part 2. Consider subdural hematoma as traumatic and keep the code as indexed, even though the manner of death is selected as "Disease".

- 3b) when above does not apply and the condition is reported in Part 2:

keep the code as indexed, regardless of the manner of death.

Example 4

- 1 (a) Pneumonia [CA40.Z](#)
(b)
(c)
(d)
- 2 Subdural hematoma [NA07.6Z](#)

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) which indicates a traumatic origin. There is no cause reported in a sequence that would indicate its origin, and there is no external cause reported. Keep the code as indexed, even though the manner of death was selected as "Disease".

3c) When above does not apply:

code to the origin indicated by the manner of death. if the manner of death is "Pending Investigation, or Unknown (not selected)", keep the code as indexed.

Example 5

- 1 (a) Subdural hematoma {[NA07.6Z]} [8B02](#)
(b)
(c)
(d)
- 2

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) which indicates a traumatic origin. However, there is no cause reported in a sequence, no external cause reported, and the only information indicating its origin is the manner of death. Assign the code following the manner of death and code to [8B02](#) *Nontraumatic subdural haemorrhage*.

Additional examples:

Example 6

- 1 (a) Subdural hematoma {[NA07.6Z]} [8B02](#)
(b) Cerebrovascular accident [8B20](#)
(c)
(d)
- 2

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic* which indicates a traumatic origin. There is another condition 'Cerebrovascular accident' indexed to a nontraumatic origin, reported as causing subdural

hematoma. Consider subdural hematoma was nontraumatic, and code to [8B02](#) *Nontraumatic subdural haemorrhage*.

Example 7

- 1
 - (a) Subdural hematoma {[NA07.6Z]} [8B02](#)
 - (b) Cerebrovascular accident [8B20](#)
 - (c)
 - (d)
- 2

Manner of death: Accident

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic*. The condition is reported as due to a nontraumatic condition in Chapter 8. Consider subdural hematoma as nontraumatic, and code to [8B02](#) *Nontraumatic subdural haemorrhage*, even though the manner of death is selected as “Accident”.

Example 8

- 1
 - (a) Subdural hematoma, dementia [NA07.6Z,6D8Z](#)
 - (b)
 - (c)
 - (d)
- 2

Manner of death: Accident Subdural hematoma is indexed to [NA07.6Z](#) which indicates a traumatic origin. A nontraumatic condition, dementia is reported together but is not in a sequence. The manner of death indicates that it was a traumatic origin. Consider subdural hematoma as traumatic, and code to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic* as indexed.

Example 9

- 1
 - (a) Intracerebral haemorrhage [8B00.Z](#)
 - (b) Meningioma, Subdural hematoma {[NA07.6Z]} [2A01.0Z,8B02](#)
 - (c)
 - (d)
- 2

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic* which indicates a traumatic origin. There is no cause reported as causing subdural hematoma, and there is no external cause reported on the certificate. The only information indicating the origin is the manner of death. Assign the code following the manner of death and code to [8B02](#) *Nontraumatic subdural haemorrhage*.

Example 10

1 (a) Subdural hematoma [NA07.6Z](#)

(b)

(c)

(d)

2

Manner of death: Disease

Description of external cause: Fell in hospital

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic* which indicates a traumatic origin. There is no cause reported in a sequence, but an external cause is reported in Frame B. Consider subdural hematoma as traumatic and keep the code as indexed, even though the manner of death is selected as "Disease".

Example 11

1 (a) Acute heart failure [BD1Z](#)
[&XT5R](#)

(b) Subdural hematoma [8B02](#)
{[NA07.6Z]}

(c)

(d)

2 Cerebral aneurysm, Fell while walking 3 months, 1 year
[8B22.5], [PA6Z](#)

Manner of death: Disease

Subdural hematoma is indexed to [NA07.6Z](#) which indicates a traumatic origin. There is no cause reported as causing subdural hematoma. An external cause, Fell while walking is reported in Part 2 but the duration indicates it was a different event from the subdural hematoma. The only information indicating the origin is the manner of death. Assign the code following the manner of death and code to [8B02](#).

Example 12

1 (a) Subdural hematoma [NA07.6Z](#)

(b)

(c)

(d)

2 Hip fracture [NC72.2Z]

Manner of death: Disease

Description of external cause: Fell in hospital

Subdural hematoma is indexed to [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic* which indicates a traumatic origin. There is no cause reported in a sequence, but an external cause is reported in Frame B. There is another injury in Part 2 but this does not necessarily mean subdural hematoma was unrelated to the external cause reported. Consider subdural hematoma as traumatic and keep the code as indexed even though the manner of death is selected as “disease”.

2.22 Mortality digital end to end solution (forms, tools and training modules)

electronic Medical Certificate of Cause of Death (eMCCD)

The medical certificate of cause of death is the basis for collecting information on causes of death. The form is designed to collect all relevant aspects to assigning the cause of death and is independent of the revision of ICD. In addition to the paper form template in section [3.14](#), the technical specifications for a digital form are made available. The specifications serve to standardize input in line with the standard form and include a data dictionary for field names and content-encoding that allows processing the content with the API for coding, as well as with software for selection of the single underlying cause of death.

Digital tools for the Cause of Death

Coding software for mortality consists of programs that assist in coding, selecting the single underlying cause of death and assessing the quality of the coding or the quality of the data at the population level.

The WHO Digital Open Rule Integrated Cause of Death Selection (DORIS) software assists in the selection of the underlying cause of death selection. It can be used online and offline. The ICD mortality rules detailed in section [2.20](#) were converted into a digital format for processing by this software. WHO is aiming at convergence in the different efforts in programming such software.

Analysing Mortality and Causes of Death 3 (ANACOD-3)

This software performs a comprehensive and systematic analysis of mortality and cause-of-death data. ANACoD3 analyzes sub-national level data to inform of potential health equity issues or outbreak patterns; it also analyzes data over multiple time periods for trend analyses and allows for the analysis of cause-of-death data coded in ICD-10 as well as ICD-11 formats. It is available in multiple languages. For assistance in using the tool, please send an email to mortality@who.int.

CodEdit

Software that helps producers of cause-of-death statistics in strengthening their capacity to perform routine plausibility checks on their coding of data. Allows for the analysis of cause-of-death data coded in ICD-10 as well as ICD-11 formats

WHO Interactive self-learning tool

The WHO electronic training tool is designed for self-learning and classroom use. The modular structure of this training permits user groups specific tailoring of courses on individual paths if desired. There are two mortality related modules of the training tool.

Certification self-learning module

A training module using the cause of death certificate version for persons that fill in causes of death on a death certificate.

Recommendations for conducting an external inspection of a body module and how to fill in a death certificate

A document providing recommendations for conducting an external inspection of a body and filling in the medical certificate of cause of death (MCCD) using the WHO 2016 international medical certificate of cause of death.

Other software or tools can be accessed through
<https://www.who.int/standards/classifications> or <https://icd.who.int>

2.23 Main uses of the ICD: Morbidity

Morbidity data are used for statistical reporting mostly at national or local levels. While some of this statistical reporting is conducted within an academic research context, it is commonly conducted in applied settings to inform health system and public health agency decision-making. ICD coded data also forms the basis of different casemix systems such as different varieties of Diagnosis Related Groups (DRGs). Coded morbidity data can also be used to inform a variety of clinical guidelines through provision of foundational information on burden of disease. The rules given here are primarily for international reporting and analysing purposes but are also recommended as a standard for national use.

2.23.1 ICD use in clinical care

Clinical care comprises different levels of treatment, all of which mean a level of diagnostic capacity that is higher than in primary care. The ICD addresses this level of detail primarily through multidimensional coding. Secondary care refers to the health care services provided by medical specialists and other health professionals who generally do not have first contact with patients, for example, cardiologists, urologists, or dermatologists. It includes acute care, necessary treatment for a short period of time for a brief but serious illness, injury or other health condition, such as in a hospital emergency department. It also includes skilled attendants during childbirth, intensive care, and medical imaging services. 'Secondary care' is sometimes used synonymously with 'hospital care'. However, many secondary care providers do not necessarily work in hospitals, such as psychiatrists, clinical psychologists, or physiotherapists, and some primary care services are delivered within hospitals. Depending on the organisation and policies of the national health system, patients may be required to see a primary care provider for a referral before they can access secondary care. Tertiary care refers to specialised consultative health care, usually for inpatients and following a referral from a primary or secondary health professional, in a facility that has personnel and facilities for advanced medical investigation and treatment, such as a tertiary referral hospital.

2.23.2 ICD use for epidemiological purposes

Epidemiology is the study of the distribution and determinants of health-related states or events (including disease) and the application of this study to the control of diseases and other health problems. Various methods can be used to carry out epidemiological

investigations- surveillance and descriptive studies are used to study distribution and analytical studies are used to study determinants. ICD coded data, either from morbidity or mortality sources, contribute to the understanding of the health of a population.

2.23.3 ICD use in quality and patient safety

Coded health information is used to measure and report on various aspects of quality of care and patient safety (e.g. reporting on in-hospital mortality or adverse event rates for various conditions or reporting on patient safety indicators). Users of this kind of health information are health system payers (e.g. ministries of health, or in privately-funded health care systems, health insurance companies) and other stakeholders, such as health quality councils, hospital administrators, clinical leaders/groups, or public advocacy organisations.

2.23.3.1 The quality and safety use case for ICD-11

The quality and safety use case of the ICD is based on the availability of large numbers of methodological tools that are originally based on ICD-10. Specific examples include the Charlson and Elixhauser co-morbidity indices, AHRQ (Agency for Healthcare Research and Quality) Patient Safety Indicators, the Hospital Standardised Mortality Ratio, and various other administrative data quality indicators. WHO recommendations on coding rules for hospital separation episodes improve comparability of records across hospitals and jurisdictions. Specific examples of coding rules include: a) rules for specifying the main condition, b) numbers of codes per record, c) code clustering mechanisms, and d) use of a status display system that distinguishes diagnoses arising during a hospital stay from those present at admission. Quality and patient safety reporting is often focused not only on diagnostic information available in the International Classification of Diseases, but also on procedure information, that is currently coded in various country-specific procedure coding systems. The harmonisation of ontological concepts in international procedure coding systems will be important going forward. The available medical and surgical complication codes of ICD-11 are in line with current knowledge in the domain of safety and adverse events.

2.23.3.2 Reporting on indicators of quality of care and patient safety

This use case relates to the use of coded health information to measure and report on various aspects of quality of care and patient safety (e.g. reporting on in-hospital mortality or adverse event rates for various conditions, or reporting on patient safety indicators). The initiating actor may be a health quality council, hospital administrators, clinical leaders/groups, a health system payer (e.g. ministries of health, or in privately-funded health care systems, health insurance companies) or a public advocacy 'watch-dog' organisations. The participating actors are hospital administrators, clinicians, health system decision makers, public representatives, patients and their families, and sometimes even the media. Requirements for reporting of quality and safety data are: - Availability of person-level data on episodes of health care delivery (e.g. hospitalisations, physician visits) - Identifiers that permit attribution of the health care delivery episode to a provider, provider group, or a given health facility/hospital. - Clinical information on diagnoses present and procedures performed during a health care delivery episode. - Clinical information on relevant outcomes such as mortality, length of stay, and specific adverse events. - Analytical expertise among initiating actors so that attention is paid to data validity considerations,

knowledge of 'best' indicators (e.g. the most valid patient safety indicators), risk adjustment methodology, etc.

The outputs are reports containing information on dimensions of system quality. These can either provide global information on system performance, or comparative information stratified by provider unit (e.g. physician-level, hospital-level, or regional reporting).

2.23.3.3 Functionality:

An ideal course of events for such use would include: - The initiating actor communicates desire to conduct quality/safety measurement and reporting to relevant stakeholders. - Appropriate applications are made to secure access to the data needed to conduct the planned reporting. - Appropriate methodological and clinical expertise is enlisted to ensure that best methodological practices are incorporated into the planned reporting, and that clinical face validity and acceptability are considered. - Data analysis and reporting are undertaken. - Broad dissemination and knowledge translation to stakeholders is undertaken. A continuous quality improvement process is undertaken in response to reports (with consideration given to quality improvement interventions, and repeat measurement after intervention). - Exceptions: Quality/safety reporting that does not follow the sequence of steps described above can be compromised. Indeed, there are many historical instances of failed or suboptimal quality/safety reporting from administrative data because of skipped steps. (e.g. 1. quality reporting without valid indicators, or appropriate methodologies for risk adjustment, 2. quality reporting without good clinical face validity, 3. quality reporting without a Continuous Quality Improvement (CQI) mind set, etc.). - Examples of sub-use cases (addressing the quality dimensions of effectiveness, efficiency, safety, access) - reporting on global mortality by facility (e.g. the hospital standardised mortality ratio - HSMR) - reporting on condition-specific mortality - reporting on patient safety indicators - reporting on global or condition-specific length of stay - reporting on readmission rates after hospitalisation - reporting on global or condition-specific costs of care (e.g. cost per hospitalisation) - reporting on waiting times - reporting on small area variability in utilisation

2.23.3.4 Additional information:

Requirements:

See 'Requirements' section above. There needs to be ongoing development and refinement of quality reporting methodologies (in essence, ongoing research around the development of new administrative data quality indicators and new methodologies for risk adjustment in outcome/quality reporting).

Assumptions:

An underlying assumption in quality or patient safety reporting from administrative data is uniformity of data format and data validity across comparator units (i.e. across provides, hospitals, or jurisdictions). Uniformity in data format and validity is not always present and has been a common reason for criticism of quality or patient safety reports derived from administrative data. In this regard, WHO is promoting ongoing efforts towards standardised coding rules help to facilitate comparative reporting by reducing data variability across comparator units (e.g. rules on factors such as the definition of the 'main condition',

numbers of possible codes per record, and the implementation of diagnosis-timing codes). See also the separate use case description for international comparative reporting.

In countries with well developed health information systems, it is quite typical for episodes of hospital-based care to be recorded with varying numbers of ICD codes describing all medical conditions/diagnoses that impacted an episode of care. Furthermore, some systems allow for distinction between diagnoses that were present at time of admission versus diagnoses that arose de novo during the hospital stay. In such systems, the resulting “discharge abstract” can be used for comprehensive and routine morbidity reporting, within which quality and safety concepts are a core component. These include: - adverse events causally linked to medical care - problems in the care process (e.g. error in administration of drug), but without actual adverse event - new diagnoses arising during a hospital stay where causal link to medical care is uncertain

To record and analyse the occurrence of such events, it is crucial for health information systems to have routine morbidity reporting, within which quality and safety concepts are an important component of morbidity coding.

2.23.3.5 Recommendations for use and interpretation of coded data

These recommendations apply to the use of records in which data were captured and organised as recommended in the previous section: - Select records involving a quality or patient safety event: these are all records with any quality or patient safety harm code. - Summarise types of quality or patient safety harm represented in a set of records: select records with any quality or patient safety harm code. - Summarise the distribution of quality or patient safety Harm codes present in the selected set. - Summarise quality or patient safety causes of harm in a set of records. - Summarise quality or patient safety mechanisms in a set of records. - Summarise quality or patient safety harm in a set of records.

2.23.3.6 ICD use for research purposes

The morbidity use case for ICD–11 includes a number of situations where the primary goal is to work in an academic research paradigm to extract information from ICD–11 coded data to study burden of disease, clusters of disease, geographic distribution of diseases, and health impacts associated with various diseases. The research paradigm is of course most relevant when it has translational relevance to either health system policy or public health policy, in which case the research paradigm, labelled as such, becomes indistinguishable from applied morbidity analyses conducted for the purposed of health planning. Explicit mention is made here of the widespread use of ICD–11 coded data in a research paradigm, recognising that this is a significant driver for developing a clinically rich and detailed classification system, with novel features and coding rules that enhance the classification’s potential as a research tool.

2.23.3.7 ICD use in primary care

Primary care has been defined as essential front line health care based on practical, scientifically sound, and socially acceptable methods and technology made universally accessible to individuals and families in the community through their full participation and at a cost that the community and the country can afford to maintain. Of relevance to primary care, ICD–11 includes many diagnostic and disease entities that are common

reasons for care at the first level of health services. This could be family practitioners, local health nurses, or in other settings also specialists, like Ophthalmologists.

ICD–11 has a simplified version for low diagnostic resource primary care settings. For high diagnostic resource settings, the tabular list for mortality and morbidity statistics contains all elements relevant to primary care and is thus able to be used in high resource environments for primary care, as well as for secondary and tertiary care.

The new version of WONCA's (World Organization of National Colleges, Academies, and Academic Associations of General Practice/Family Practitioners) International Classification of Primary Care and the ICD–11 will share a common subset of categories.

2.23.3.8 ICD use in Casemix groupings

In casemix grouping systems such as the Diagnosis Related Group (DRG) system, ICD based data are used for reimbursement and/or resource allocation. Such systems are used nationally in systematic fashion.

The assignment of patient cases to groups is based on an algorithm using, in addition to coded diagnosis information, coded procedures, and a number of other variables. The scientific basis of the casemix systems is grounded in health care economics and in the theory of medicine. Since casemix systems are an essential part of administration in countries that use them, smooth transfer to the new revision of the ICD in these systems is essential for the approval and implementation of the new revision.

ICD–11 has been developed to accommodate the different levels of detail that are required in diagnosis-related casemix groupings, in close collaboration with the custodians of the diverse casemix systems. Joint use in a specific casemix system is driven by the relevant grouper algorithms, and partly also by national legislation.

For matters of international comparability of hospital activity, it is recommended that countries adopt the new WHO definition of main diagnosis and country implementations of ICD–11 apply the new extension codes for the types of diagnoses that are provided with ICD–11.

For international tabulations, in view of the differences in definitions of hospital, episode and in-patient, the resulting diagnoses are listed with the aid of the International Shortlist for Hospital Morbidity Tabulation (refer to section [2.25](#)).

2.23.4 What is coded: Patient conditions

2.23.4.1 Main condition

The definition of main condition relates to the description of an episode of hospital-based care.

The health care practitioner should record and identify as the main condition the one condition that is determined to be the reason for admission, established at the end of the episode of health care. This determination is supported through evaluations and investigations that aim to establish the diagnosis responsible for the admission.

2.23.4.2 Multiple conditions contributing to need for admission

Where an episode of health care includes more than one condition contributing to the need for admission (e.g. congestive heart failure and pneumonia; acute cerebral haemorrhage and hip fracture; multiple injuries - concussion, rib fracture, right femur fracture after MVA; or influenza A and Type 1 diabetic ketoacidosis), the health care practitioner should record and identify the main condition to be the one condition that is deemed to be the most clinically significant reason for admission.

2.23.4.3 Other conditions

In addition to the main condition, the health care practitioner should, whenever possible, also list separately all other conditions or problems dealt with during the episode of health care. Other conditions (also known as additional diagnoses) are defined as those conditions that coexist at the time of admission or develop during the episode of health care and affect the management of the patient. Conditions related to an earlier episode that have no bearing on the current episode should not be recorded as other conditions. It is recommended, where practicable, to carry out multiple-condition coding and analysis to supplement the routine data.

2.23.5 Health care practitioner documentation guidelines for morbidity coding

The health care practitioner responsible for the patient's treatment is also responsible for documenting the patient's health conditions. This information should be organised systematically by using standard recording methods. A properly completed medical record is essential for good patient management. It is also an essential prerequisite to the creation of a valid coded record of patient diagnoses, derived through a coding process from written information describing a patient's medical condition. When a good written record of patient conditions is available, successful coding of this information in ICD and associated classifications produces a valuable source of epidemiological and other statistical data on morbidity and other health care problems.

The person transforming the information on the stated condition to codes (the 'coder') may be the health care practitioner or a clinical coder who is not responsible for the patient's treatment but is specially trained in use of the classification to assign the code(s). In the latter situation, which is quite common among member countries, the coder depends on the adequacy of clinical documentation relating to the patient's condition(s) provided by health care practitioners in the medical record. The primary importance of clinical documentation by health care practitioners as the starting point for coded health data cannot be overstated and needs to be underlined as being a matter of key importance within countries and internationally – with implications for health information and clinical documentation teaching within health care practitioner training programs.

For clinical and resource allocation purposes, in many instances, the manifestation of a disease (kind and severity, e.g. ulcer grade 3) may be more relevant during a specific treatment episode than the underlying disease (e.g. Diabetes mellitus). For prevention programs at national levels, knowledge about the underlying aetiology may be more important. Quality and safety assessments will require reporting additional detail related to the stay. For comprehensive analysis and use of morbidity data, it is crucial to have a dataset with multiple fields allowing capture of codes relating to all the aspects above.

Morbidity data are increasingly being used in the formulation of health policies and programmes, and in their management, monitoring and evaluation, in epidemiology, in identification of at risk populations, and in clinical research (including studies of disease occurrence in different socioeconomic groups).

In the context of these morbidity coding rules, the term practitioner is used throughout the morbidity rules to mean physician or any qualified health care practitioner who is legally accountable for establishing the patient's diagnosis. This information should be organised systematically by using standard recording methods. A properly completed record is essential for good patient management and is a valuable source of epidemiological and other statistical data on morbidity and other healthcare problems.

The term episode is used for all settings, including hospital admissions. It is acknowledged that the definition may be different in each country, though it is most often considered to be a continuous hospital care period, which begins on the first day of a patient's admission to a health care facility and ends on the day upon which they are separated from that facility through discharge, transfer or death. Some countries consider sequential care periods on different wards within the same hospital to be distinct episodes of care.

The health care practitioner responsible for the patient's treatment is also responsible for documenting the patient's health conditions during an episode of health care. Good clinical documentation is critical to continuity and quality of patient care, communication between members of the treatment team, patient safety and is the legal record of a patient's episode of care. When a sound written record of patient condition(s) is available, successful coding of this information using the International Classification of Diseases (ICD) and associated classifications produces a valuable source of morbidity data to support:

- Health care planning, management, monitoring and evaluation
- Epidemiology
- Identification of risk populations
- Clinical research
- Reimbursement and health care funding.

The health care practitioner responsible for the patient's treatment should select and document the main condition, as well as any other conditions, for each episode of health care. It is recommended, where practicable, that all conditions are documented to support multiple condition coding and analysis to supplement routine data collection and reporting.

2.23.5.1 Documentation guidelines involving the term 'Multiple'-For Single condition reporting

In cases involving, for example, 'multiple fractures', 'multiple head injuries' or 'multiple valvular disease', it is acceptable documentation practice to record the diagnoses using the term 'multiple' and then list separately the specific conditions or injuries. For example, Multiple fractures of pelvis: fracture of os pubis, sacrum, ilium.

2.23.5.2 Specificity and detail

Each diagnostic statement should be as informative as possible in order for the clinical coder to classify the condition to condition to a code that best captures the specific detail provided in the diagnostic statement. Examples of such diagnostic statements include:

- transitional cell carcinoma of trigone of bladder
- acute appendicitis with localised peritonitis
- meningococcal pericarditis
- pregnancy-induced hypertension
- diplopia due to reaction to antihistamine taken as prescribed
- osteoarthritis of hip due to an old hip fracture
- fracture of neck of femur following a fall at home
- full thickness burn of palm of left hand due to grilling accident
- unintentional puncture of the sigmoid colon during colonoscopy

2.23.5.3 Unconfirmed diagnoses

If no definite diagnosis has been established at the end of an episode of health care, then the health care practitioner should document the information that permits the greatest degree of specificity and knowledge about the reason for admission that has been established at the end of the episode of care. This could be a symptom, abnormal finding or problem. Rather than qualifying a diagnosis as “possible”, or “suspected”, when a diagnosis has been considered but not established, when applicable, record the symptom, abnormal finding or problem.

2.23.5.4 Documentation of a ruled out condition

The health care practitioner should document as main condition a “ruled out” condition when the episode of care involves a person who presents some symptoms or evidence of an abnormal condition which requires study, but who, after examination and observation, show no need for further treatment, follow-up or other medical care.

The health care practitioner should not document a ruled out condition as a main condition if some treatment was provided for a symptom or follow-up is required to determine the cause of the sign or symptom. In that instance, the health care practitioner should document the presenting sign or symptom that was treated as the main condition.

Example 1

Admitted for suspected deep vein thrombosis of leg, which after investigation is ruled out and no follow-up necessary.

Main Condition: Ruled out deep vein thrombosis.

Example 2

A child is found playing with an empty acetaminophen bottle. The mother is uncertain if there were any tablets in the bottle. The child is brought to the hospital and following investigation, it is determined that the child did not ingest any pills

Main condition: Ruled out unintentional ingestion acetaminophen (paracetamol)

Example 3

A patient had an elevated PSA and presented for biopsy of the prostate. Biopsy revealed no evidence of malignancy. No further follow-up planned for the patient.

Main condition: Ruled out prostate malignancy

See Section [2.23.17 Coding a “ruled out” condition](#) for more detail.

2.23.5.5 Contact with health services for reasons other than illness

Episodes of health care or contact with health services are not restricted to identification, treatment or investigation of current illness or injury. Episodes may also occur when someone who may not currently be sick requires or receives limited care or services. In this case the health care practitioner should document the details of the relevant circumstances as the ‘main condition’.

Examples include:

- monitoring of previously treated conditions
- immunisation
- contraceptive management, antenatal and postpartum care
- surveillance of persons at risk because of personal or family history
- examinations of healthy persons, e.g. for insurance or occupational reasons
- seeking of health-related advice
- requests for advice by persons with social problems
- consultation on behalf of a third party
- organ or tissue donors
- circumstances related to drugs, procedures, or devices without documented injury or harm to patient

Chapter 24 ‘Factors influencing health status and contact with health services’ provides a broad range of categories for classifying these circumstances. Reference to this chapter will give an indication of the detail required to permit coding to the most relevant category.

2.23.5.6 Conditions due to external causes

When a condition such as an injury, poisoning or other effect of an external causes is recorded, it is important to document fully both the nature of the condition and the circumstances that gave rise to it. For example:

- ‘fracture of neck of femur caused by fall due to slipping on pavement’
- ‘cerebral contusion caused when patient lost control of car, which hit a tree’
- ‘unintentional poisoning, patient drank disinfectant in mistake for soft drink’
- ‘severe hypothermia, patient fell in her garden in cold weather’

See also Section [2.23.20.2 Causation in the context of quality and safety](#).

2.23.5.7 Documentation of sequelae

Where an episode of care is for the treatment or investigation of a residual condition (sequela) of a disease that is itself no longer present, the health care practitioner should document the residual condition (sequela) and its origin, together with a clear indication that the original disease is no longer present. For example:

- 'deflected nasal septum– fracture of nose in childhood'
- 'contracture of Achilles tendon – late effect of injury to tendon'
- 'infertility due to tubal occlusion from old tuberculosis'.

Where multiple sequelae are present and treatment or investigation is not directed predominantly at one of them, a documented statement such as 'sequelae of cerebrovascular accident' or 'sequelae of multiple fractures' is acceptable.

2.23.6 Coder guidelines for selecting 'main condition' and 'other conditions' for coding purposes

The main condition and other condition(s) relevant to an episode of health care should have been identified and recorded by the responsible health care practitioner, and coding will therefore usually be straightforward. The main condition recorded should be accepted for coding and reporting unless it is obvious that the health care practitioner did not follow the guidelines for recording diagnostic information for morbidity data analysis. Whenever possible, a record with an obviously inconsistent or incorrectly recorded main condition should be returned to the health care practitioner for clarification.

If clarification of potentially erroneous documentation is not possible, one of the following rules can be applied by the clinical coder and the main condition reselected for reporting purposes. The rules are for use when it is unclear which recorded condition should be selected as the main condition for reporting purposes.

- MB1 Several conditions recorded as 'main condition'; or
- MB2 Condition recorded as 'main condition' is presenting symptom of diagnosed, treated condition; or
- MB3 Signs and symptoms recorded as 'main condition' with alternative conditions recorded as the cause

Coder rules for reselection of main condition

2.23.6.1 MB1 - Several conditions recorded as 'main condition'

If several different conditions (that cannot be classified to a single stem code) are recorded as the 'main condition', and other details on the record point to one of them being the 'main condition' (one condition determined to be the reason for admission established at the end of the episode of care), select that condition; otherwise, select the condition first recorded.

If there is the desire to also report other discharge diagnosis types i.e. main resource condition or initial reason for encounter or admission, then the applicable extension code(s) from Chapter X 'Extension codes', should be assigned to indicate the different types of discharge diagnosis types that are reported.

Example 1:

A patient was admitted with complaints of fever, chills, severe headache and stiff neck. Following investigation, a diagnosis of staphylococcal meningitis was confirmed. While in hospital the patient developed pneumonia.

Main condition: Staphylococcal meningitis. Pneumonia

Two conditions have been recorded as the main condition and querying the health care practitioner is not possible. The details in the example point to staphylococcal meningitis as the one condition being the reason for admission established at the end of the episode of care, therefore the coder should code staphylococcal meningitis as the 'main condition'. Pneumonia is coded as an 'other condition'. It meets the definition of an 'other condition' as it is a diagnosis that arose during the episode of care.

Example 2:

A patient who has a history of COPD was admitted for a biopsy of the prostate. Patient was evaluated for COPD. Biopsy was performed and the final diagnosis from pathology results was benign prostatic hypertrophy.

Main condition: Chronic obstructive pulmonary disease (COPD). Hypertrophy of prostate.

Two conditions have been recorded as the main condition and querying the health care practitioner is not possible. The details in the example point to benign prostatic hypertrophy as the condition being the reason for admission established at the end of the episode of care; therefore, the coder should code hypertrophy of prostate as the 'main condition'. COPD is coded as an 'other condition' as the physician documented it as co-existing at the time of admission and affecting the management of the patient.

Example 3:

A patient presents to hospital at 35 weeks gestation with spontaneous premature rupture of membranes. She is not having any contractions. Examination reveals the baby is in breech presentation; therefore, delivery by caesarean section is recommended. Mother delivers healthy preterm infant by caesarean section.

Main condition: Premature rupture of membranes. Breech presentation.

Procedure: Delivery by caesarean section

Two conditions have been recorded as the main condition and querying the health care practitioner is not possible. The details in the example point to premature rupture of membranes as the condition being the reason for admission established at the end of the episode of care. Therefore, the coder should code premature rupture of membranes as the 'main condition' and breech presentation and preterm delivery as 'other condition'.

Example 4:

A patient is admitted to the hospital with pneumonia and congestive heart failure.

Main condition: Pneumonia and Congestive Heart Failure.

Two conditions have been recorded as the main condition and querying the health care practitioner is not possible. There are no other details on the record to point to one of the conditions as being the main condition, therefore, in this instance, the coder should code the first listed condition as the main condition. Pneumonia is coded as the 'main condition' and congestive heart failure is coded as an 'other condition'.

2.23.6.2 MB2 - Condition recorded as 'main condition' is presenting symptom of diagnosed, treated condition

If a symptom or sign (usually classifiable to Chapter 21 Symptoms, signs or clinical findings, not elsewhere classified), or a problem classifiable to Chapter 24 Factors influencing health status or contact with health services, is recorded as the 'main condition', and this is obviously the presenting sign, symptom or problem of a diagnosed condition recorded elsewhere and care was given for the latter, reselect the diagnosed condition as the 'main condition'.

Example 1:

The patient presents to hospital with complaint of haematuria. Investigations reveal a papilloma in the posterior wall of the bladder as the cause of the haematuria. The papilloma was excised by diathermy.

Main condition: Haematuria

Other conditions: Papilloma of posterior wall of bladder

Haematuria (symptom) is recorded as the main condition; however, it was determined to be caused by the papilloma of the bladder (diagnosed and treated condition). Therefore, the coder should reselect and code papilloma of posterior wall of bladder as the 'main condition'.

Example 2:

The patient presents to hospital with abdominal pain. Investigations reveal acute appendicitis and the patient undergoes an appendectomy.

Main condition: Abdominal pain

Other conditions: Acute appendicitis

The symptom 'abdominal pain' was recorded as the main condition; however, it was determined to be caused by appendicitis. Therefore, the coder should reselect and code acute appendicitis as the 'main condition'.

Example 3:

A patient with known COPD is admitted to hospital with acute respiratory failure which after investigation is found to be caused by an acute exacerbation of COPD.

Main condition: Acute respiratory failure

Other condition: Acute exacerbation of COPD

The symptom “acute respiratory failure” was recorded as the main condition; however, it was determined to be caused by exacerbation of COPD with acute exacerbation. Therefore, the coder should reselect and code the COPD as the ‘main condition’

2.23.6.3 MB3 - Signs and symptoms recorded as ‘main condition’ with alternative conditions recorded as the cause

Where a symptom or sign is recorded as the ‘main condition’ with documentation that it may be due to either one condition or another, select the symptom as the ‘main condition’.

Example 1:

Main condition: Headache due to tension or acute sinusitis

The symptom ‘headache’ is recorded as the main condition with two possible causes; therefore, the coder should code headache as the ‘main condition’.

2.23.7 Coding using postcoordination in morbidity

A significant new feature in ICD–11 is an embedded functionality for postcoordinating detail related to diagnostic concepts see Section [2.10.2 Combining stem codes and extension codes, and how to order these in a complex code cluster](#), and Section [2.10.1 Adding detail – postcoordination and cluster coding with multiple stem codes and extension codes](#).

The postcoordinated coding of diagnostic concepts described by the health care practitioner is shown in the following examples:

Example 1

A patient is admitted to the hospital for laser treatment of their diabetic retinopathy due to Type 2 diabetes mellitus. During the admission the patient’s medication for arterial hypertension required adjustment on a number of occasions before discharge. Code as main condition the diabetic retinopathy, unspecified postcoordinated with the stem code type 2 diabetes mellitus ([9B71.0Z](#)/[5A11](#)). Code the other condition, essential hypertension ([BA00.Z](#)).

For morbidity coding, the order of the codes in the first cluster in Example 1 has the diabetic retinopathy ordered first as it is the diabetic retinopathy that meets the definition of main condition followed by the causing condition Type 2 diabetes. (Note: The classification instructs to code also the type of diabetes.)

Where an established causal relationship is not documented or cannot be inferred, the two stem codes cannot be part of the same cluster.

Example 2

Patient admitted for right cataract extraction. The patient also has Type 2 diabetes mellitus and was reviewed by the endocrinologist and dietitian for their long-term diet and insulin plan. Code the main condition as Cataract, unspecified, right ([9B10](#)&[XK9K](#)). Code the other condition as Type 2 diabetes mellitus ([5A11](#)).

Example 2 demonstrates postcoordination where a causal relationship between the cataract and the Type 2 diabetes has not been documented and cannot be inferred; therefore, the two stem codes for each condition are reported separately. Postcoordination for the laterality for the cataract was applied here.

2.23.7.1 Coder rule for use of extension codes

Adding detail to Stem codes using Extension Codes

Type 2 extension codes (a new section of codes in ICD–11) provide distinct codes that serve as concept modifying flags for marking how a diagnosis is to be used and/or interpreted.

Examples of these extension code modifiers include:

- Discharge diagnosis types (main condition; main resource condition; initial reason for encounter or admission);
- Diagnosis certainty (Provisional diagnosis; Differential diagnosis)
- Diagnosis Timing (Present on admission; Developed after admission; Uncertain timing of onset relative to admission)

For more detail about the use of all available extension codes, see Section [2.9Extension codes](#).

Example 1

A patient is admitted to hospital with chest pain and after investigation is diagnosed with a myocardial infarction and then develops a stroke that leads to a one-month hospitalisation. Myocardial infarction is coded as the main condition because it was the reason for admission established at the end of the stay. The stroke is coded separately and can be postcoordinated with a diagnosis-type extension code flag indicating that the stroke diagnosis arose after admission.

Such a system with diagnosis flags meets the objectives of countries that want a reason-for-admission coding rule, while also meeting the objectives of countries that want to be able to make inferences regarding complications of care and resource consumption (which are of relevance to casemix systems and patient safety and quality assessments).

2.23.8 Coding from health care practitioner documentation of ‘causal relationships’

Sometimes conditions that have a causal relationship are clearly documented by the health care practitioner using terms such as ‘due to’, ‘caused by’, or ‘arising from’. These connecting terms indicate the health care practitioner has made a causal link between, for example, condition A due to condition B. However, sometimes conditions are documented with connecting terms that are ambiguous for the coder such as ‘with’, ‘after’, ‘in’, and ‘following’. When ambiguous terms are documented, and it is not clear whether the health care practitioner means a causal relationship or not, the clinical coder should code each condition separately and not link the conditions in a cluster.

The clustering (postcoordination) is a particularly notable new feature in ICD-11 that has permitted the introduction of powerful new clinical coding mechanisms for capturing clinical information in dimensions such as:

- quality and safety coding for health care related injury and harms (see three-part model described in Section [2.23.20.1](#))
- injury and the external cause of injury
- the addition of clinical detail using extension codes
- the specification of diagnosis type and diagnosis timing using extension codes
- the comprehensive description of late effects (sequelae) arising from prior conditions (See Section [2.21.6](#))
- the description of inter related stem code diagnoses where there is a clear causal relationship between them

For more information on causal inference in the context of quality and safety, refer to Section [2.23.20.2 Causation in the context of quality and safety](#).

2.23.9 Coding of suspected conditions or symptoms, abnormal findings and non-illness situations

If the episode of health care was for an inpatient, the coder should be cautious about classifying the main condition to Chapters 21 Symptoms, signs or clinical findings, not elsewhere classified and Chapter 24 Factors influencing health status or contact with health services. If a more specific diagnosis has not been made by the end of the inpatient stay, or if there was truly no codable current illness or injury, then codes from the above chapters are permissible. The categories can be used in the normal way for other episodes of contact with health services.

If, after an episode of health care, the main condition is recorded as ‘suspected’, ‘questionable’, etc., and there is no further information or clarification, the suspected diagnosis must be coded as if established.

Example 1

Main condition: Suspected acute cholecystitis If there is no further information available that indicates that a definitive diagnosis was reached, code to acute cholecystitis, unspecified ([DC12.0Z](#)) as ‘main condition’.

Example 2

Main condition: Severe epistaxis. Patient in hospital one day. No procedures or investigations reported. Code to epistaxis ([MD20](#)). Although epistaxis is a sign/symptom, it is acceptable since the patient was obviously admitted to deal with the immediate emergency only.

2.23.10 Coding using combination categories

The ICD provides certain categories where two conditions or a condition and an associated secondary process can be represented by a single code (i.e. precoordinated concept). Such combination categories should be used where appropriate information is recorded.

Example 1:

Main condition: Chronic Kidney Disease (CKD), Stage 4 secondary to hypertensive renal disease Other condition: Essential hypertension Code to Chronic Kidney disease, stage 4 ([GB61.4](#)) and add the Hypertensive renal disease ([BA02](#)) in a cluster. Main condition Cluster: [GB61.4/BA02](#) Other condition: [BA00](#) *Essential hypertension*

Example 2:

Main condition: Glaucoma secondary to eye inflammation Code to [9C61.24](#) *Glaucoma due to eye inflammation*. This is a precoordinated code in ICD-11.

Example 3:

Main condition: Diabetic cataract. Type 1 diabetes mellitus Other conditions: Hypertension Code Diabetic cataract ([9B10.21](#)) and 'code also' the type of diabetes mellitus ([5A10](#)). Postcoordinate the stem code for diabetic cataract and stem code for Type 1 diabetes mellitus. Main condition Cluster: [9B10.21/5A10](#). The hypertension is not linked to the cataract or diabetes, so it is not part of the cluster. It is coded as an 'other condition'.

Example 4:

Main condition: Rheumatoid arthritis Other conditions: Hypertension, Type 2 diabetes mellitus, Cataract Code to Rheumatoid arthritis, serology unspecified ([FA20.Z](#)) as 'main condition'. Code the other conditions (hypertension ([BA00.Z](#)), type 2 diabetes mellitus ([5A11](#)), cataract ([9B10.Z](#))) separately. If any optional extension codes are added to any one of the conditions, then a cluster(s) is created as applicable because extension codes cannot be reported alone. Note that in this example, the linkage (through clustering) of cataract with diabetes must not be made as the cataract has not been documented as a diabetic cataract. In this case there is no combination indicating clustering should be used.

Main condition: [FA20.Z](#) Other condition: [BA00.Z](#) Other condition: [5A11](#) Other condition: [9B10.Z](#)

2.23.11 Coding using external causes of morbidity

For injuries and other conditions due to external causes, both the nature of the condition and the circumstances of the external cause should be coded. The preferred 'main condition' code should be that describing the nature of the condition. This will often, but not always, be classifiable to Chapter 22 Injury, poisoning or certain other consequences of external causes. The code from Chapter 23 External causes of morbidity or mortality indicating the external cause is assigned as an additional code and postcoordinated with the nature of the condition as it may be regarded as a modifier. See also Section [2.23.20.1](#).

Example 1:

Main condition: Fracture of neck of femur caused by fall Other conditions: Contusions to elbow and upper arm The health care practitioner has identified the fracture as the main condition and since there is no other information to make the coder question the main condition recorded, the coder should code fracture of neck of femur, unspecified ([NC72.2Z](#)) as the 'main condition'. The external cause code for unintentional fall from unspecified height ([PA6Z](#)) is used as an additional code linked to the fracture code through postcoordination. The contusion of elbow ([NC30.1](#)) and upper arm ([NC10.1](#)) are coded as another condition cluster and the external cause code for unintentional fall ([PA6Z](#)) is linked to the contusion code through postcoordination.

Main condition cluster: [NC72.2Z/PA6Z](#) Other condition cluster: [NC30/NC30.1/NC10.1](#)

Example 2:

Main condition: Hip fracture from fall Other condition: Severe hypothermia resulting from exposure to the cold weather Code to [NC72.2Z](#) *Fracture of neck of femur, unspecified* as 'main condition' and postcoordinate the external cause code for [PA6Z](#) *Unintentional fall from unspecified height*. Code as "other condition" hypothermia ([NF02](#)) and postcoordinate the external cause code [PB16](#). Main condition cluster: [NC72.2Z/PA6Z](#) Other condition cluster: [NF02/PB16](#)

Example 3:

Main condition: Diplopia due to reaction to antihistamine taken as prescribed Code to diplopia [9D46](#) and postcoordinate the external cause code for [PL00](#) *Drugs, medicaments or biological substances associated*

with injury or harm in therapeutic use and [PL13.2](#) *Drug-related injury or harm in the context of correct administration or dosage, as mode of injury or harm*. An optional extension code may be added to identify the specific drug was an antihistamine XM4J58.

Main condition cluster: [9D46/PL00&XM4J58/PL13.2](#)

2.23.12 Coding of acute and chronic conditions recorded as main condition

Where the main condition is recorded as being both acute (or subacute) and chronic, and the ICD provides separate categories or subcategories for each, but not for the combination, the code for the acute condition should be reported as the main condition (the condition determined to be the reason for admission established at the end of the episode of care). When an appropriate combination code is provided for both the acute and chronic condition, assign the combination code as the main condition.

Example 1:

Main condition: Acute on chronic cholecystitis Code [DC12.00](#) *Acute on chronic cholecystitis* as the 'main condition'. This is an example of combination code for both the acute and chronic condition in ICD-11.

Example 2:

Main condition: Acute exacerbation of chronic obstructive pulmonary disease Code to [CA22.0](#) *Chronic obstructive pulmonary disease with acute exacerbation, unspecified* as the 'main condition' since the ICD provides an appropriate single precoordinated code for the combination.

2.23.13 Coding of injuries or harm arising from surgical or medical care

Refer to Section [2.23.20.1 Overview of code-set in ICD–11 for quality and patient safety](#).

2.23.14 Coding of adverse events and circumstances in health care that do not cause actual injury or harm

Refer to Section [2.23.20.1 Overview of code-set in ICD–11 for quality and patient safety](#).

2.23.15 Coding of chronic postprocedural conditions

Most body-system chapters also contain categories for permanent (chronic) conditions that occur either as a consequence of specific procedures and techniques or as a result of the removal of an organ, e.g. postmastectomy lymphoedema syndrome, post-irradiation hypothyroidism. Immediate or acute conditions that occur as a consequence of a procedure may require coding with the 3-part quality and safety model. See also Section [2.23.20.1 Overview of code-set in ICD–11 for quality and patient safety](#). The categories of postprocedural conditions do not have residual codes (i.e. other and unspecified). This is intentional so as to prevent users inadvertently classifying conditions to these categories that should, in fact, be classified elsewhere.

2.23.16 Coding 'History of' and 'Family history of'

Chapter 24 'Factors influencing health status or contact with health services' of the classification includes a number of codes that describe both a personal history of various conditions, and a family history of various conditions. The classification of this documented concept may be coded in one of two ways.

Option 1: Assign the applicable stem code from Chapter 24 ‘history of’ (or ‘family history of’) by itself.

Option 2: Assign the applicable stem code from Chapter 24 clustered with a code from another chapter to add specificity as to what the previous ‘disease’ was. The order of stem codes in the cluster is always the ‘history of’ stem code first, followed by any other codes that may be added for detail.

Example 1:

A patient has history of sigmoid colon cancer that was curatively resected. Code: [QC40.0](#) Personal history of malignant neoplasm of digestive organs/[2B90.3Z](#) Malignant neoplasm of sigmoid colon, unspecified

In example 1, it is acceptable to code only [QC40.0](#) as the code simply captures the notion that the patient has a personal history of cancer of the digestive organs. However, the documented clinical concept (history sigmoid colon cancer) is fully described through use of the clustering mechanism and linking of stem codes [QC40.0/2B90.3Z](#).

Example 2:

A patient has a family history of macular degeneration. Code: [QC66](#) Family history of eye or ear disorders/[9B78.3Z](#) Degeneration of macula or posterior pole, unspecified

2.23.17 Coding a “ruled out” condition

Many health care encounters are undertaken to evaluate patients for suspected conditions, to then determine after investigations that the patient does not have the condition in question. In medical documentation, such scenarios often use the term “ruled out”. It is essential for health information systems to have an ability to report on such encounters.

ICD-11 includes a number of codes that can be used to describe encounters where a suspected condition has been “ruled out”. These appear in Chapter 24 as children of [QA02](#) *Medical observation or evaluation for suspected diseases or conditions, ruled out*. Some of these codes specify the suspected condition in question:

- Observation for suspected malignant neoplasm, ruled out
- Observation for suspected tuberculosis, ruled out
- Observation for suspected allergy or hypersensitivity, ruled out

In many other common scenarios, there is no code for a specified suspected condition, in which case, [QA02.Y](#) *Medical observation or evaluation for other suspected diseases or conditions, ruled out*, is used. In such cases, postcoordination can be used to specify the suspected condition that was ruled out. The classification of the documented concept “ruled out” may be coded in one of two ways. Option 1: Assign the applicable code from [QA02](#) *Medical observation or evaluation for suspected diseases or conditions, ruled out*, by itself. Option 2: Assign the applicable code from [QA02](#) *Medical observation or evaluation for suspected diseases or conditions, ruled out*, clustered with a code from another chapter to add specificity as to what was the suspected disease that was ruled out. The order of stem codes in the cluster is always the [QA02](#) stem code first.

Example 1